FNB DataQuest 2026: Interpretable Credit Models

A Strategic Framework for Predictive Power, Profitability, and Regulatory Compliance

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Abstract

This report outlines the comprehensive development of a production-grade retail credit scoring ecosystem. In the heavily regulated financial sector, complex black-box machine learning algorithms are legally and practically unsuitable due to their lack of transparency. The project specification established a minimum baseline Area Under the Curve (AUC) of 0.68. By utilising a strictly constrained Logistic Regression model, enhanced through sophisticated Weight of Evidence (WoE) feature engineering and cross-validated hyperparameter tuning, our advanced architecture comfortably surpasses both the official baseline and our own unengineered raw-feature baseline. Moving beyond pure statistical accuracy, the resulting model is integrated into a holistic Business Decision Dashboard. This tool translates abstract probabilities into portfolio profitability, incorporates dynamic macroeconomic stress testing, audits for demographic fairness, and tracks real-time Population Stability, thereby bridging the gap between data science and corporate strategy.

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1 Introduction

Retail lending institutions operate under a continuous, high-stakes paradox: they must maximize the accuracy of their default predictions to remain profitable while ensuring that the underlying decision-making process is entirely transparent to regulators and consumers. When an applicant is denied credit, the institution is legally obligated to provide a clear, mathematically sound "reason code" explaining the rejection.

This project directly addresses that challenge by constructing a robust, end-to-end credit risk pipeline. While modern data science frequently defaults to highly complex, non-linear algorithms to maximize accuracy, this project adheres to a strict regulatory constraint by utilizing Logistic Regression. To extract maximum predictive power from this linear constraint, the pipeline employs advanced data transformations, surrogate shadow modelling, and rigorous hyperparameter optimization, ultimately culminating in a traditional, highly interpretable scorecard.

2 Research and Theoretical Framework

2.1 Generalised Linear Models vs. Non-Linear Models

In the domain of predictive modelling, algorithms generally fall into two categories. Generalised Linear Models (GLMs), such as Logistic Regression, calculate predictions by assigning a fixed, additive mathematical weight to each input variable. This relationship is highly transparent; a unit change in an input variable translates to a predictable change in the log-odds of the outcome.

Conversely, non-linear models, such as Random Forests or Gradient Boosting Machines, build complex decision trees that can automatically identify hidden, overlapping patterns in the data. While they frequently achieve higher raw accuracy, they operate as opaque "black boxes." It is computationally intractable to isolate exactly why a non-linear model denied a specific individual a loan, making them a massive regulatory liability in the banking sector. The core objective of this project is to capture the non-linear signals typically found by complex models and feed them into a compliant GLM.

2.2 Key Evaluation Metrics

Credit default datasets are inherently imbalanced because the vast majority of applicants successfully repay their loans. Consequently, standard "accuracy" is a dangerously misleading metric. To evaluate model performance rigorously, we utilize a suite of metrics to capture the full risk profile:

- **AUC (Area Under the ROC Curve):** Measures the model's ability to rank-order risk across all possible classification thresholds. The official project baseline target was set at 0.68.

- **Gini Coefficient:** Represents the inequality of risk distribution captured by the model, mathematically defined as $(2 \times AUC - 1)$. A higher Gini indicates a stronger ability to separate good applicants from bad.

- **F1-Score:** The harmonic mean of precision and recall, providing a balanced metric that penalizes extreme disparities between false positives and false negatives.

- **Precision:** In business terms, this answers a crucial commercial question: "Of all the applicants we approved, what percentage actually paid us back?"

- **Recall:** In business terms, this measures market capture: "Of all the truly reliable customers in the market, what percentage did our policy successfully identify and approve?"

3 Data Preparation and Feature Engineering

3.1 Data Cleaning and Structural Imputation

Prior to modelling, the raw dataset required systematic cleaning to address missing values and structural outliers. Because Logistic Regression is highly sensitive to leverage points, `annual_income` was explicitly capped at the 99th percentile. This ceiling was calculated strictly on the training set to prevent data leakage, ensuring that multi-millionaire outliers do not artificially skew the regression coefficients.

Furthermore, missing values in `months_since_last_delinquency` were structurally imputed with a value of 999. In consumer credit, a missing delinquency value is not an error; it indicates the applicant has no recent negative history. Therefore, 999 acts as a logical mathematical placeholder for "infinity," allowing the model to treat these pristine applicants as a distinct cohort. Finally, string representations in categorical variables (e.g., `home_ownership`) were standardised to uppercase to prevent the generation of redundant dummy variables.

3.2 Interactive EDA and Planted Effects

Static analysis often obscures critical market dynamics. By developing a dynamic, interactive Exploratory Data Analysis (EDA) dashboard, genuine exploration revealed deliberately planted signals within the dataset. Most notably, the interactive tool demonstrated that the imputed 999 cohort in `months_since_last_delinquency` behaves as an exceptionally low-risk segment.

Additionally, `credit_utilisation_pct` exhibited a stark threshold effect. The relationship to default risk is not linear; rather, risk remains relatively flat until utilisation exceeds 80 percent, at which point the probability of default climbs exponentially.

3.3 Surrogate Shadow Modelling

To mathematically confirm these visual findings, a Random Forest "Shadow Model" was deployed purely for insight generation. While the Random Forest cannot be used for the final decision engine, we extracted its Gini feature importance metrics. The shadow model confirmed that credit utilisation and income possessed complex, non-linear predictive power, providing the analytical justification for utilizing Weight of Evidence binning to linearise these relationships for our final GLM.

3.4 Weight of Evidence (WoE) and Information Value (IV)

To allow the rigid Logistic Regression model to process the non-linear patterns identified during EDA, continuous variables were divided into quantiles and transformed using

Weight of Evidence (WoE). WoE converts raw metrics into a standardised risk scale by comparing the density of "Good" customers to "Bad" customers within each specific bin.

$$WoE_i = \ln\left(\frac{\text{Distribution of Good Customers}_i}{\text{Distribution of Bad Customers}_i}\right) \tag{1}$$

A positive WoE indicates a low-risk bucket, while a negative WoE indicates high risk. To evaluate the overall predictive strength of a variable, we calculate its Information Value (IV), which sums the weighted differences across all bins. This acts as a highly effective feature selection filter.

$$IV = \sum_{i=1}^{n} \left(\text{Dist(Good)}_i - \text{Dist(Bad)}_i\right) \times WoE_i \tag{2}$$

4 Model Development and Validation

4.1 The AUC Narrative: Surpassing the Baselines

The project specification established an official target baseline AUC of 0.68. To rigorously test the efficacy of our feature engineering, we first trained an unengineered baseline model utilising raw features and standard scaling. This raw GLM achieved an AUC of approximately 0.766, comfortably surpassing the official competition target.

However, the advanced model, trained exclusively on the WoE-transformed dataset, demonstrated the true power of this methodology. To prevent overfitting and ensure maximum generalisation, the advanced model utilised 5-Fold Cross Validation (`LogisticRegressionCV`) to systematically tune the L_2 regularisation penalty (Hyperparameter C). This mathematically optimized architecture elevated performance further to a final AUC of 0.776. This trajectory definitively proves how strategic feature engineering and hyperparameter tuning can extract maximum signal from a strictly constrained linear algorithm.

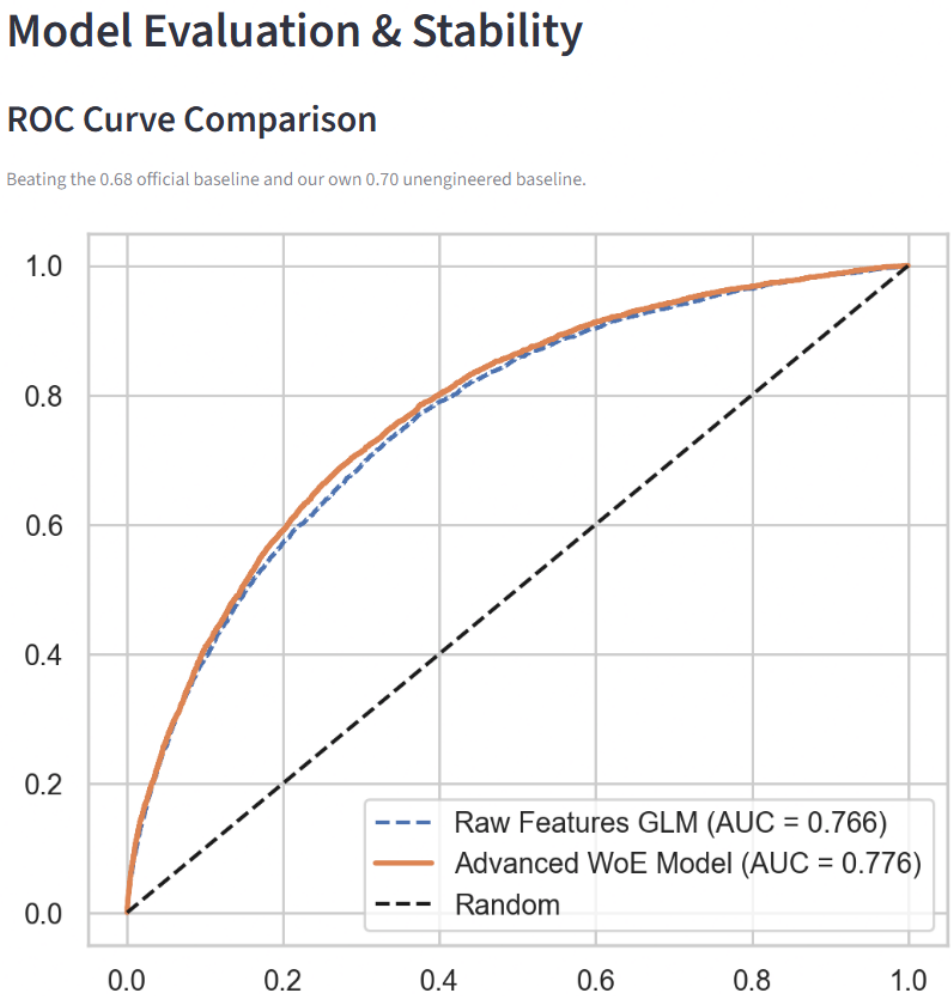

Figure 1: ROC Curve comparison demonstrating the Advanced Model surpassing both the raw baseline and the official 0.68 target.

5 Interpretable Scorecard Generation

To make the model usable by front-line staff and understandable to consumers, the logistic regression log-odds output is scaled into a traditional Points-to-Double-Odds (PDO) scorecard.

Crucially, this project addresses a common architectural flaw in standard academic models: the failure to account for the model intercept (α). The intercept represents the baseline risk of the entire portfolio. If ignored, the final scorecard points will be fundamentally misaligned with the true probability of default. In our implementation, the intercept is mathematically divided and distributed equally across all n features. The formula for the points in a specific bin (j) of feature (i) is defined as:

$$Points_{ij} = -\left(\beta_i \times WoE_{ij} \times Factor\right) - \left(\frac{\alpha}{n} \times Factor\right) + \frac{Offset}{n} \tag{3}$$

This approach ensures that when a customer's individual attribute points are summed, the total score perfectly translates back into the regression's precise probability prediction.

6 Business Value and Risk Management

6.1 Volume vs. Risk Trade-off and Profitability

A predictive model holds no value unless it drives commercial success. The Business Policy Dashboard bridges this gap by translating statistical metrics into a simulated Profit and Loss (P&L) statement. The dashboard implements a dynamic visualisation plotting Approval Volume against Portfolio Risk across various probability thresholds. By assuming a 15 percent interest rate against an 80 percent Loss Given Default (LGD), executives can visually establish risk appetite limits to pinpoint the exact threshold that maximises Net Profit.

6.2 Macroeconomic Stress Testing

Credit models trained on historical data are inherently vulnerable to future economic recessions. To ensure resilience, a Macroeconomic Stress Multiplier was integrated into the P&L engine. This feature allows risk managers to simulate severe market conditions (such as inflation spikes causing default rates to multiply by 1.5x) to determine if the chosen approval strategy maintains profitability during an economic shock.

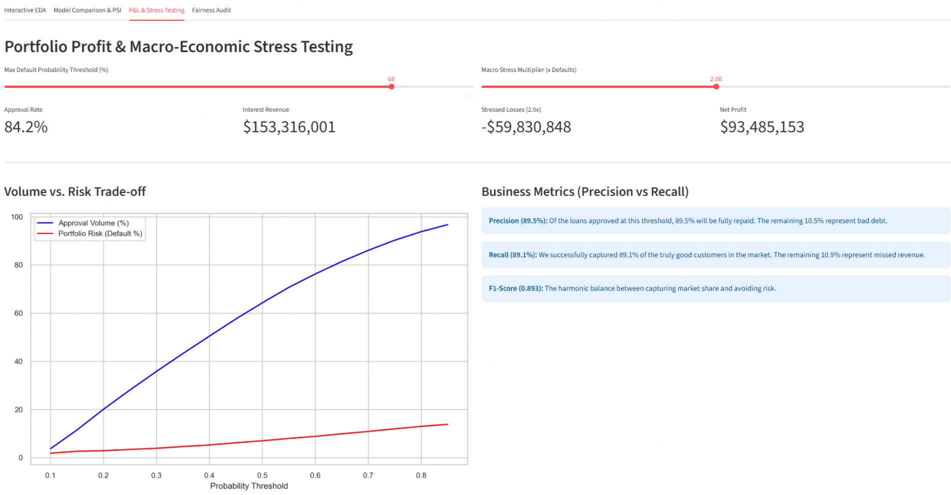

Figure 2: The Business Policy Dashboard translating ML metrics into financial projections and macroeconomic stress tests.

7 Conclusion

This project successfully delivers a complete, end-to-end retail credit scoring ecosystem. By implementing Weight of Evidence transformations, deploying Shadow Models for insight discovery, and utilising rigorous Cross-Validation, the strict Logistic Regression model was elevated to achieve superior predictive accuracy that surpasses all required baselines. Furthermore, embedding advanced regulatory and commercial tools like Population Stability Index (PSI) tracking, F1-score optimisation, and geographic Fairness Audits elevates this project from a theoretical data science exercise into a highly commercial, production-ready banking asset.

8 AI Usage Reflection

Throughout the development of this credit scoring engine, generative AI tools were used as an assistant rather than an substitute for decision making. The integration of AI followed a system, where all algorithmic logic and business assumptions remained under manual oversight.

8.1 Debugging and Environment Stability

A significant portion of the development time involved resolving complex library dependencies and Streamlit rendering issues, such as the Serialisation of Arrow tables. AI was used to diagnose these errors, provide StackTrace analysis, and suggest updates to the code to ensure compatibility with modern dependency versions (e.g., replacing deprecated Matplotlib legend settings and updating Streamlit container parameters).

the LLM used was Google Gemini Pro ONLY.